



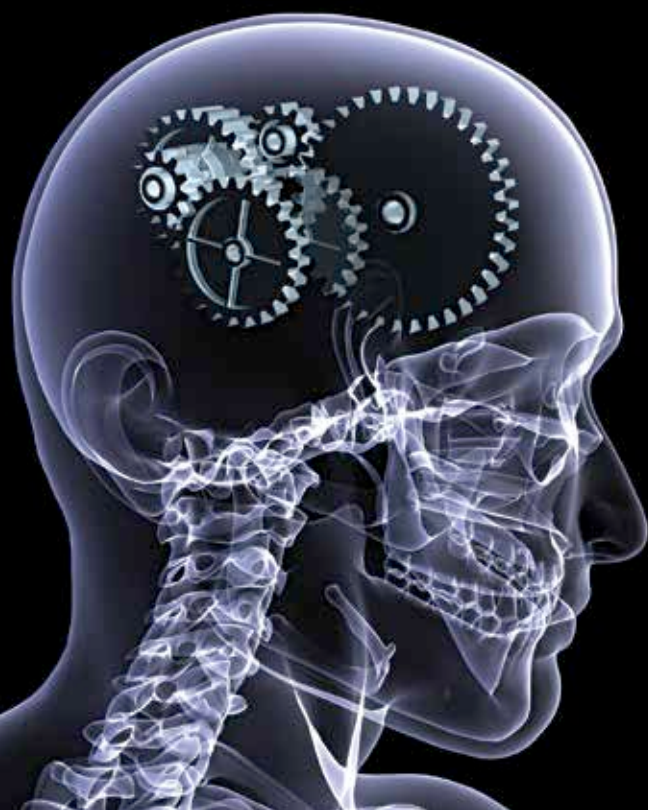
Traces of oldest stars found

Astronomers have found the remains of a star that is a 100 times the size of the sun and possibly the oldest in the universe <http://dgit.in/OldestStars>



Peck, Peck, Peck!

Ever wondered why Woodpeckers don't get brain damage? Here's why: <http://dgit.in/WoodPeck>



Thinking Machine

The Future of Computing With Neuromorphic Technology

Scientists developing neuromorphic computing are creating machines that aren't just faster but truly intelligent. What's the future of humanity when your computer's brain is smarter than yours?

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The human brain is the most complex machine in existence. What's most unusual is its disproportionately large size considering the size of the average human because the brain is twice as large in humans as it would be in any other mammal our size. The brain has been considered the factory of thought, a pen-sive for experience and the citadel of the soul. The brain has evolved over a period of seven million years and given birth to consciousness, yet we still have no clue how it works.

Human brains are like biological computer processors that are capable of amazing performance. Technologies like predictive analysis and robotics have, for the longest time, attempted to mimic the human thought process, but have remained bound by the linear dimensions of simple logic and basic mathematics. The mind has always remained a challenge, an elegant machine that is more than just a calculator, with a vast potential for innovation, imagination and learning,

which machines have yet to even attempt replicating. Until now.

Brain Built to Order

As spring emerged in 2014 San Diego, a baby sized robot called Pioneer was busy learning pattern recognition in Qualcomm's headquarters. The robot was successfully able to navigate the test area, which simulated a child's room, and was able to seek, identify and store toys that were strewn about all over the place - making decisions quickly without having any programmed instructions teaching it to distinguish toys from other things. The robot was instructed where to store the toys by observing human hand gestures



The Pioneer robot learnt correct behaviour through positive feedback for successfully performed tasks.

and interpreting their meaning to discern toys from everything else.

Even though Pioneer was just a carrier of arduously complicated programming code, it proved that neuromorphic principles can radically enhance machine intelligence. The incorporation of brain-based computing in silicon chips has been one of the upcoming thresholds of computing for a long time now. It isn't surprising that the granddaddy of modern computing - IBM - has been the one to push it to its limits beating out Qualcomm with a giant leap forward.

With the August 2014 release of their IBM SyNAPSE chip they produced the first ever production scale neuromorphic, or brain like, processor chip. This chip - called TrueNorth - is the result of IBM's deal with the United State's Defense Advanced Research Project Agency (the folks who invented the internet, a.k.a., DARPA). As a part of DARPA's Systems of Neuromorphic Adaptive Plastic Scalable Electronics or SyNAPSE program, this chip has been six years in development with collaboration from Cornell Tech and iniLabs costing over \$50 million.

TrueNorth is the most mature neuro-chip or neuromorphic chip in existence